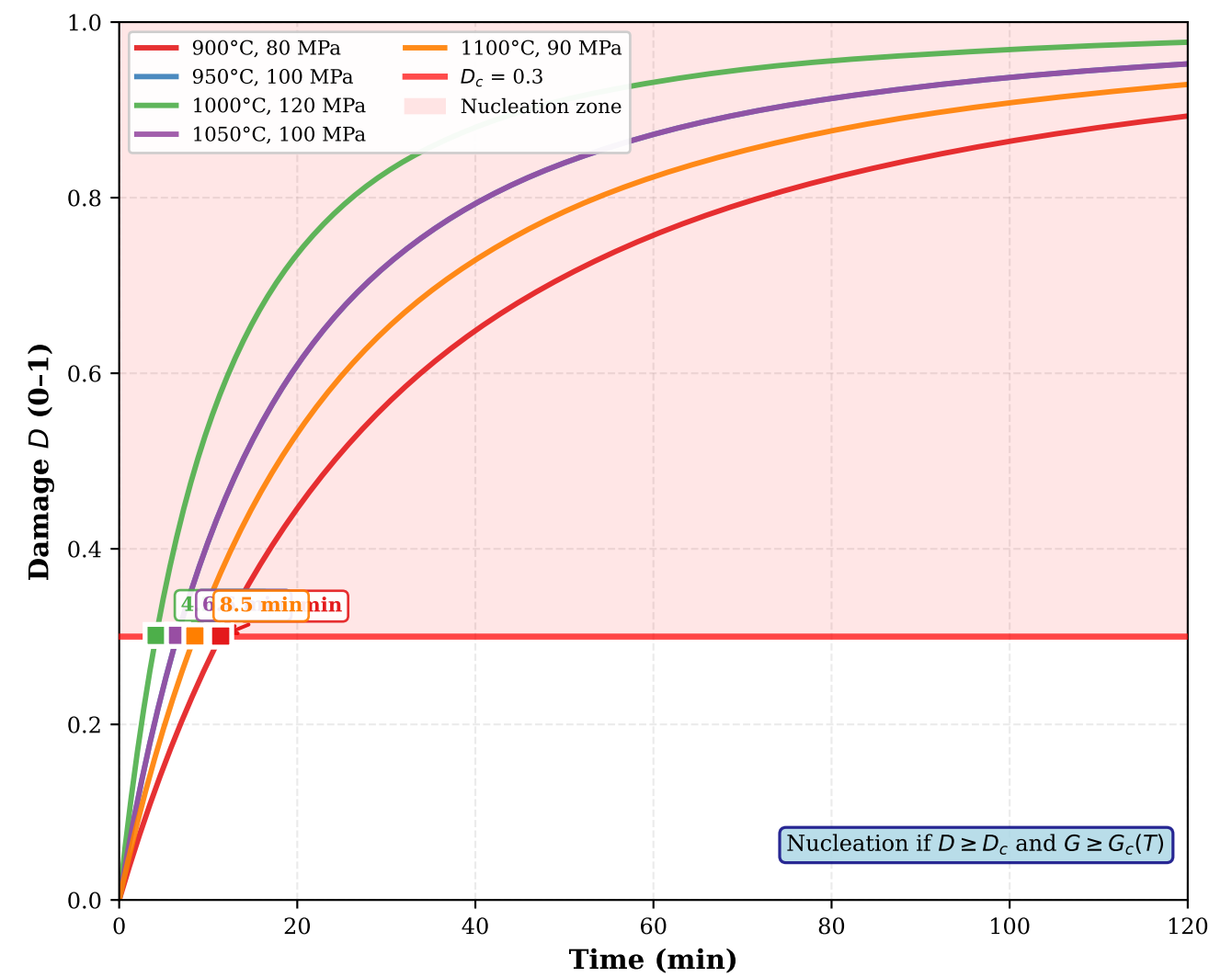
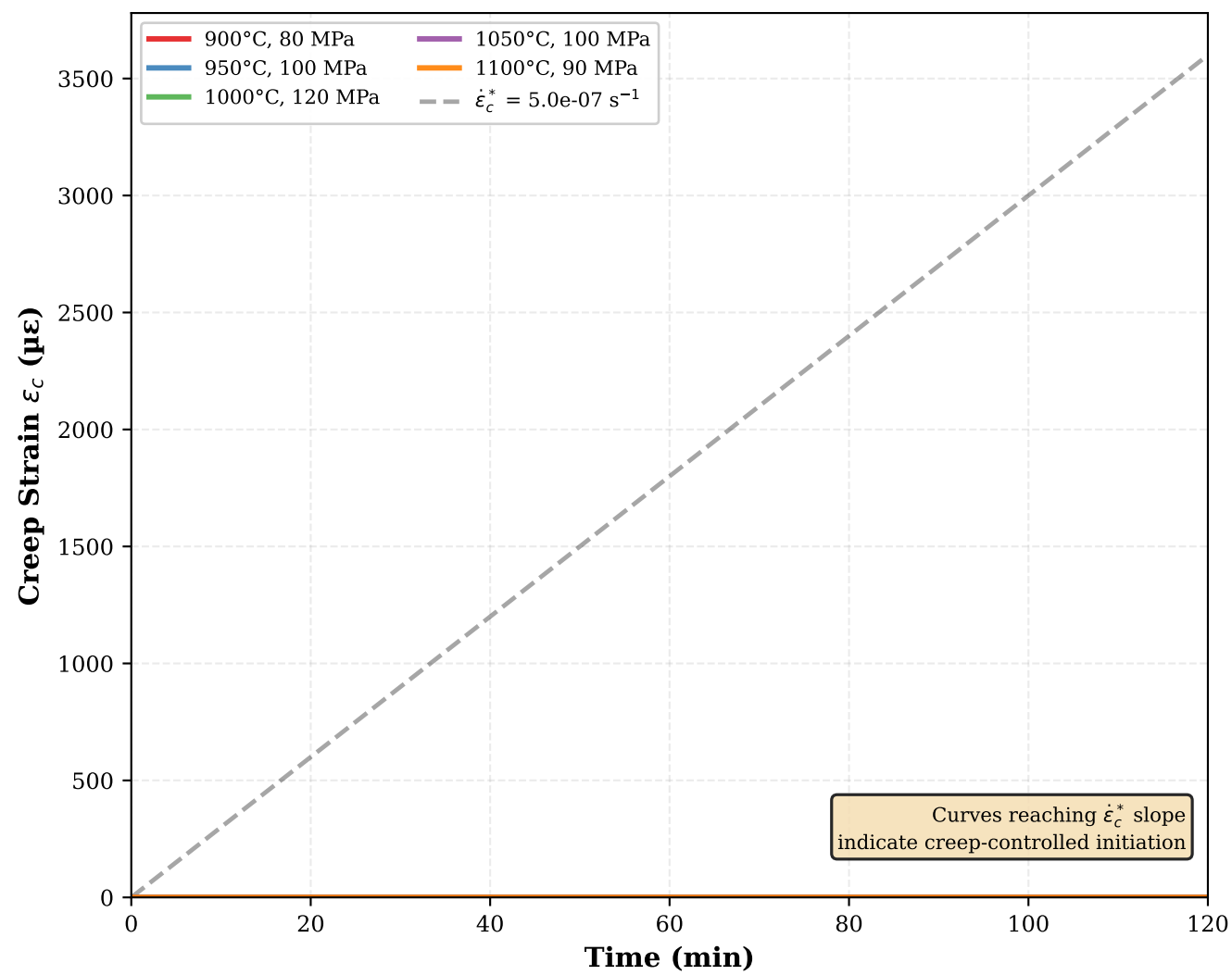
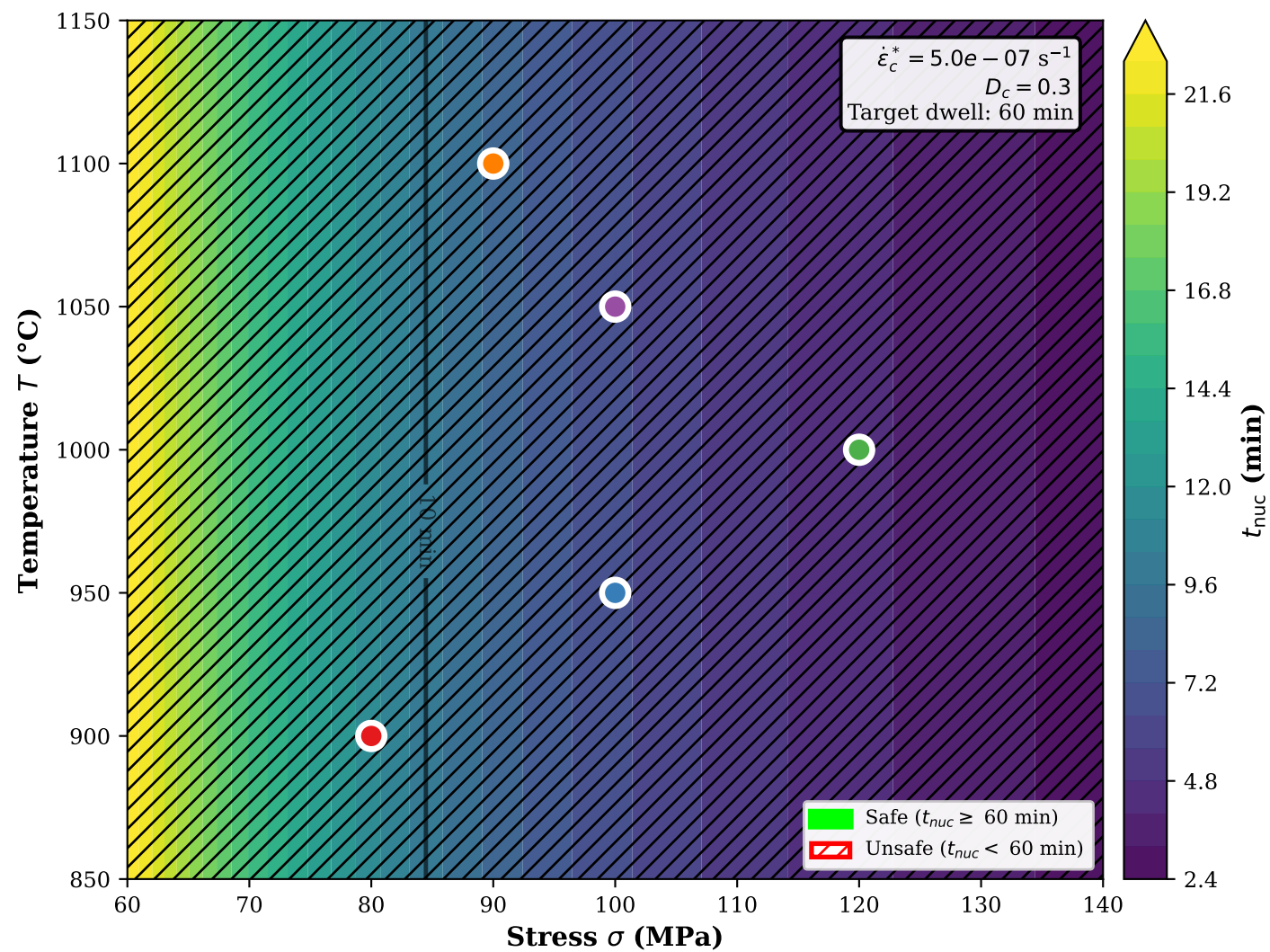


**Figure 4a.2: Creep Thresholds and Microcrack Initiation—Multi-Panel Synthesis**  
**Panel A: Creep Strain Evolution and Threshold Detection** **Panel B: Damage Accumulation and Nucleation Onset**



**Panel C:  $\sigma$ - $T$  Hazard Map and Operating Envelope**



**Panel D: Experimental Validation (DIC and XRD)**

